

Model fixes and the GP / embedding comparison

Five fixes found while comparing the paper's embedding systems, and what the comparison shows

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1 Summary

- Five defects were found and fixed while setting up the comparison. Fixes 1, 2, 4 and 5 were also present in the code the paper's hardware flights used (tag `archive/main-2026-05-02`), and in the code of the paper's SITL runs.
- **The GP decides whether the certificate is true when the wind drifts; the embedding system barely matters here.** In the numerical runs (the paper's evolving wind) the time-varying GP keeps the vehicle inside its certified tube with every embedding system (0 % of the time outside); a time-invariant GP is violated about 40 % of the time, by up to 0.29 m. The SITL world has no wind, so SITL cannot show this.
- With the time-varying GP the three embedding systems certify the same horizon with the same tubes; (66)-(67) is the cheapest (about a third less compute per rollout), so **TV-GPR + (66)-(67) is the default (the baseline)**. Rolled out from identical inputs the tubes are not nested as the construction suggests ((68) (66) (64)); see "Are the tubes nested?" below.
- In SITL a **5 cm model-mismatch margin** on δ (now the default) takes the time the vehicle spends outside its certified tube from 19 % to 0 % (all six variants: 0-0.02 %), at the price of a shorter certified horizon (0.34 s).

2 The five fixes

	what was wrong	effect	fix
1	the embedding used $+ [M_u]K$, but the controller applies $u = u_r - K(x - x_r)$	the tube modelled the wrong closed loop (looser than needed)	$K_p = -K$ in (66)-(69); certifies 0.45-0.62 s instead of 0.40-0.53 s on the same plans
2	the dynamics' Jacobian was taken over $w \in \mu(x_r) \pm s$ only	the $\partial/\partial\theta$ entries depend on w ; the domain must contain every disturbance the box can produce	domain $[\underline{\gamma}, \bar{\gamma}]$ over the box, for all variants
3	<code>RolloutPlan.index_at</code> truncated exact row times ($0.03/0.01 = 2.999... \rightarrow 2$)	45 % of numerical ticks used the previous row (controller and metrics)	$+10^{-6}$ row tolerance (Python and C++)
4	the planar state's h, v were filled with inertial (v_y, v_z); the model defines them in the body frame	up to 1.7 m/s off at 30° roll, in the tube's start state and the RTA feedback	$[h; v] = R(\theta)^\top [v_y; v_z]$ (Python and C++)
5	the state came from the <code>mocap_px4_relays</code> relay: a 25 ms timer behind a 50 Hz rate gate that local position (exactly 50 Hz) keeps failing	25 Hz with gaps up to 150 ms, i.e. stale state	the node and the C++ loop read <code>/fmu/out/vehicle_odometry</code> (100 Hz) directly

Effect of fixes 4 and 5 on a default SITL flight:

	before	+ body-frame h, v	+ 100 Hz PX4 odometry
tracking RMSE lateral / vertical	88 / 67 mm	28 / 43 mm	20 / 32 mm
worst position outside the certified tube	0.45 m	0.21 m	0.11 m
C++ fast loop, RMSE lateral / vertical	86 / 50 mm		23 / 28 mm

New metric: escapes. `analysis.tube_escapes` measures, at every certified control tick, how far the vehicle's position is outside the certified box of the row in use: the true state in the numerical simulation, the EKF2 estimate in SITL and on hardware. A certificate that is correct has no escapes beyond the 10 ms integration step of the tube (about 1 mm).

The remaining SITL gap, and the model-mismatch margin. With $\delta = 0$ the TV-GPR flights still leave the tube about 20 % of the time (up to 0.12 m). The escapes happen in the **first 0.1 s of each plan** (40-46 % of those ticks, about 1 % once a plan is older than 0.4 s), while the tube, which starts from the estimated state with a ± 0.5 mm box, is millimetres wide: in its first moments the real vehicle departs from the planar model's prediction (rate-loop lag, out-of-plane motion; drag is weak, -0.08 N per m/s). The threshold does not reach this part of the tube: at 0.5 m instead of 0.25 m the time outside is 16 % (worst 0.13 m). A **model-mismatch margin** on δ does, and is now the default (`--model-mismatch-margin 0.05`: $\delta = \max(0.05 \text{ m}, \text{estimate uncertainty})$):

TV-GPR + (68)-(69), PX4 SITL, 3 flights each	$\delta = 0$	$\delta = 0.05$ m	$\delta = 0.1$ m
time outside the certified tube / worst	19 % / 0.12 m	0 % / 1 mm	0 % / 0
outside in a plan's first 0.1 s	46 %	0 %	0 %
certified horizon, median / 5th percentile	0.54 / 0.42 s	0.35 / 0.28 s	0.25 / 0.21 s
plans in 20 s	41	83	98
tracking RMSE lateral / vertical	20 / 27 mm	13 / 16 mm	10 / 11 mm

The price is a shorter horizon and twice the replanning (a rollout takes about 5 ms, so the margin to the 0.35 s horizon stays large); tracking improves partly because the reference is re-anchored at the vehicle twice as often. The numerical simulation keeps $\delta = 0$: its model is the vehicle.

3 The comparison

Two wind models (TV-GPR, $\epsilon = 0.25$; GPR = TVGPR with $\epsilon = 0$) $\times$ the paper's three embedding systems (Appendix A), all at the same certificate: position within 0.25 m of the reference, 0.3 m floor. **Numerical** closed loop: the paper's evolving wind, which **drifts over time**; exact model; 10 cases per variant (calm, $\times 0.3$ / $\times 0.6$ / $\times 1.0$ with 3 seeds), backup off, $\delta = 0$. **PX4 SITL**: 3 flights per variant, backup on, flown with $\delta = 0$ and with the default 5 cm margin. The Gazebo world has **no wind** (`default.sdf`; `windy.sdf` only blows a constant 5×2 m/s), so in SITL the GPs learn a steady thrust offset: SITL tests the model mismatch, not the wind model.

Numerical, drifting wind (9 wind runs per variant): time outside the certified tube, worst excursion:

embedding system	TV-GPR	GPR
(68)-(69): first order in u and w	A: 0 %, 0.9 mm (the paper's)	D: 41 %, 0.10 m
(66)-(67): first order in u	E: 0 %, 0.9 mm (baseline)	C: 40 %, 0.29 m
(64)-(65): no first-order terms	F: 0.03 %, 1.2 mm	B: 40 %, 0.28 m

	A	E	F	D	C	B
numerical: certified horizon (median)	0.59 s	0.59 s	0.59 s	0.78 s	0.80 s	0.70 s
numerical: tube half-width at +0.25 s	26 mm	26 mm	25 mm	2.6 mm	2.5 mm	2.4 mm
numerical: rollout	6.9 ms	4.3 ms	6.8 ms	7.0 ms	6.1 ms	6.2 ms
SITL, $\delta = 0$: time outside the tube	19 %	20 %	20 %	65 %	67 %	71 %
SITL, $\delta = 0$: worst excursion	0.12 m	0.12 m	0.15 m	0.27 m	0.80 m	0.26 m
SITL, 5 cm margin: time outside the tube	0 %	0 %	0 %	0 %	0 %	0.02 %
SITL, 5 cm margin: certified horizon	0.34 s	0.34 s	0.34 s	0.17 s	0.29 s	0.29 s

	A	E	F	D	C	B
SITL, 5 cm margin: tracking lateral / vertical (mm)	13 / 18	13 / 16	12 / 15	7 / 9	11 / 12	12 / 11

Why the GP matters (numerical). The time-invariant GP’s uncertainty stays small near its 9 observations, so its tubes are about $10\times$ thinner and certify further ahead; but the wind drifts over time, so the GP is confidently wrong and the vehicle leaves the “certified” tube about 40 % of the time, by up to 0.29 m. The TV-GPR’s forgetting grows the uncertainty over the look-ahead, which keeps the tube honest: no escapes with any embedding system.

Are the tubes nested? Every plan of the $\times 0.6$ wind run rolled out again with the three TV-GPR embeddings from its recorded inputs (34 plans):

	A (68)-(69)	E (66)-(67), baseline	F (64)-(65)
certified horizon t_r , mean	0.597 s	0.598 s	0.599 s
position half-width at +0.25 s / +0.40 s	24.4 / 82.5 mm	24.4 / 82.5 mm	24.1 / 80.8 mm
nested inside the next (all states, certified rows)	A E in 0 / 34	E F in 30 / 34	

(On 71 plans of two wind runs: $t_r(A) - t_r(E) \in [-10, 0]$ ms, $t_r(E) - t_r(F) \in [-20, 0]$ ms.) (66)-(67) bounds the GP mean by its exact range over the box; (68)-(69) replaces it by the mean-value form $[M_w][M_x^\mu](x - x_r)$, which on its own contains that range and only wins when folding it into the state matrix cancels part of $[M_x] + [M_u]K$ (a disturbance whose state dependence counteracts the dynamics, as in the paper’s example $g(x) = -x$).

Why the embedding barely matters. With TV-GPR the growth of σ along the look-ahead dominates the tube’s growth, so the first-order disturbance term of (68)-(69) buys no measurable tightening over (66)-(67); and for this vehicle (64)-(65) coincides with (66)-(67), because the inputs enter through a constant selector matrix: $[M_u](K[e]) = ([M_u]K)[e]$. A benefit of the first-order terms would need a disturbance whose mean varies across the tube much more than its uncertainty grows along it.

What SITL shows. With $\delta = 0$ every variant leaves its tube in the first moments of each plan, the thinner GPR tubes more often (65-71 % against 19-20 %); this is the planar model’s mismatch with the PX4 vehicle, not a wind-model failure (there is no drifting wind). With the 5 cm margin all six variants stay inside their tubes and none needed the LAND backup; the TV-GPR variants certify the longest, while the GPR variants replan up to $1.6\times$ as often, which re-anchors the reference and lowers their tracking error. The failure of the time-invariant GP can only be reproduced in SITL with a time-varying wind in Gazebo.

Data, tables (GP $\times$ embedding grids, two-way effect split, both SITL sets), per-flight results and figures: `packages/px4_rta_mm_gpr/scripts/data_analysis/05_embedding_comparison.ipynb`. Reproduce with `--gp tv|static --embedding uw|u|none --model-mismatch-margin 0.05` (node) or `SimConfig(gp_epsilon=..., embedding=...)`.